MIL-C-6183B
13 February 1981
SUPERSEDING
MIL-C-6183A
24 September 1963

MILITARY SPECIFICATION

CORK AND RUBBER COMPOSITION SHEET; FOR AROMATIC FUEL AND OIL RESISTANT GASKETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vulcanized sheet made of granulated cork uniformly dispersed in a synthetic rubber compound, for use in gaskets where aromatic fuel or oil resistance is required (see 6.1).

1.2 Classification. The sheet gasket material shall be of the following types, classes and grades, as specified (see 6.2):

1.2.1 Types.

Type I - Molded.
Type II - Continuous roll.

1.2.2 Classes.

Class 1 - Aromatic Fuel.
Class 2 - Oil resistant.

1.2.3 Grades.

Grade A - Soft.
Grade B - Medium.
Grade C - Firm.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 9321), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

NO INFORMATION REQUIREMENTS

FSC 5330

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-S-735	-Standard Test Fluids, Hydrocarbon.
PPP-B-576	-Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-591	-Box, Fiberboard, Wood-Cleated.
PPP-B-601	-Box, Wood, Cleated Plywood.
PPP-B-621	-Box, Wood, Nailed and Lock Corner.
PPP-B-636	-Boxes, Shipping, Fiberboard.
PPP-T-76	-Tape, Packaging, Paper (For Carton Sealing).

MILITARY

MIL-B-121	-Barrier Material, Greaseproofed, Waterproofed, Flexible.

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-Marking for Shipment and Storage.
MIL-STD-414	-Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-1188	-Commercial Packaging of Supplies and Equipment.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 395-78	-Rubber Property - Compression Set.
ASTM D 471-79	-Rubber Property - Effect of Liquids.
ASTM D 1418-79	-Rubber and Rubber Latices - Nomenclature.
ASTM D 2240-75	-Rubber Property - Durometer Hardness.
ASTM F 104-79	-Nonmetallic Gasket Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Materials. Materials used in the manufacture of the cork and rubber sheet shall be as specified below. The finished product shall have uniformly dispersed cork throughout the vulcanized rubber binder.

3.1.1 Cork. The cork shall be clean and free from hardback. The particles of cork shall be of such size that they shall pass through a 10 mesh screen and not less than 20 percent shall be retained by a 20 mesh screen.

3.1.2 Binder. The binder shall consist of a synthetic rubber type product compounded with other necessary agents to insure compliance with the requirements of this specification.

3.2 Construction. The gasket sheet material shall be fabricated by uniformly dispersing the cork in synthetic rubber and vulcanizing the resultant mixture. The methods of dispersion and vulcanization shall be as specified by the manufacturer.

3.3 Form. Unless otherwise specified (see 6.2), the gasket material shall be furnished in sheet form.

3.4 Physical properties. Physical properties shall be in accordance with table I.

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3.4.1 Resistance to immersion. Each gasket material test specimen shall show no signs of disintegration, cracking, or tackiness when visually examined after all the immersion tests specified in table I.

3.5 Dimensions and tolerances.

3.5.1 Dimensions. Unless otherwise specified, thickness and width dimensions of the gasket sheeting shall be as specified in table II. Length dimensions shall be as specified by the acquiring activity.

3.5.2 Tolerances.

3.5.2.1 Thickness. The thickness of the sheet gasket material shall not vary from the thickness specified by more than 0.015 inch (0.38 mm), and the thickness at any point on the sheet gasket material shall not vary from the thickness at any other point by more than 0.010 inch (0.25 mm), when determined in accordance with table IV.

3.5.2.2 Linear dimensions. Unless otherwise specified, width tolerances on sheet gasket material shall be 0.25 inch (6.35 mm). Length tolerances shall be ± one percent.

3.6 Finish. The finish shall be as smooth as possible giving due consideration to the coarseness of the starting materials.

3.7 Identification of product. Unless otherwise specified, all material furnished to this specification shall be legibly marked using a fuel and oil resistant lacquer, ink or dye. The marking shall not adversely affect the serviceability or performance of the gasket material. Specific markings for sheets and small parts shall be as follows:

3.7.1 Sheets. The markings on sheets (red - class 1; white - class 2) shall be 0.25 inch (6.35 mm) in height, running continuously across the sheet, recurring either lengthwise or crosswise with lines approximately 1.5 inch (38.1 mm) apart. The markings shall be on one side only and shall include the following:

Specification No.
Type, grade, class
Manufacturer's identification
ASTM D 1418 symbol for polymer used
Date of cure (1-80 indicates 1st quarter of calendar year 1980)

3.7.2 Small parts cut from sheet. When the shape and size of the material does not permit marking in accordance with 3.7.1, a green dot shall be used to indicate type I material, a yellow dot for type II, a red dot for class 1 and a white dot for class 2. For example, type I, class 1

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material shall have a green dot and a red dot in close relationship to each other. The dots shall appear on the face of the piece, if the piece has a minimum of 1 inch (25.4 mm) surface dimension in each of two directions. In the event that the dimensions of the material are such that dot marking cannot be made in accordance with the above, the individual pieces shall be stacked for marking purposes and printed identification stripes applied to the stacked edges. Where pieces are small and furnished loose in containers, markings shall appear on the containers only and shall be in accordance with that specified for sheet material.

3.8 Workmanship. The sheet gasket material shall be uniform in quality and condition, clean, smooth and free from foreign materials and defects detrimental to the appearance or performance of parts fabricated from the material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The examination and testing of the gasket sheeting shall be classified as quality conformance inspections (see 4.3).

4.3 Quality conformance inspection.

4.3.1 Lot formation. A lot shall consist of all the sheet gasket material of the same type, grade, class and thickness manufactured under essentially the same conditions and offered for inspection at one time.

4.3.2 Prior approval. When a supplier has previously delivered an acceptable product in accordance with this specification and contractual requirements, the acquiring activity may, at its discretion, waive certain requirements for subsequent acquisitions (see 4.3.3.2).

4.3.3 Sampling and inspection procedures.

4.3.3.1 Visual inspections. The sample unit for this inspection shall be one sheet or roll, as applicable. Samples shall be selected in accordance with Inspection Level II of MIL-STD-105. Each sample unit shall be inspected as specified in table III. The acceptable quality level (AQL) shall be 2.5 percent defective.

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4.3.3.2 Physical properties.

4.3.3.2.1 Initial acquisition. For first time deliveries by a contractor, sufficient gasket sheet material shall be selected from each lot to complete testing to all the properties in table IV. In addition, the contractor may be requested to furnish a quantity of sheeting in the thickness required for compression set (see 4.5.2.2). Failure of the product to pass any requirement in table IV shall be cause to reject the lot represented by the sample. Samples inspected in 4.3.3.1 may be utilized to prepare specimens for this inspection.

4.3.3.2.2 Subsequent acquisitions. The sample unit for this inspection shall be one sheet or roll, as applicable. Sample units shall be randomly selected from each lot in accordance with Inspection Level IV of MIL-STD-414 and may be selected from the samples of 4.3.3.1. Each sample unit shall be tested only to those requirements indicated by footnote 3/ to table IV. The AQL for this inspection shall be 2.5 percent defective (normal inspection) utilizing table B-3 of MIL-STD-414.

4.3.3.3 Packaging inspection. The lot size for purposes of this inspection shall be the number of shipping containers. The sample unit shall be one shipping container. Sample size shall be selected in accordance with Inspection Level S-2 of MIL-STD-105. An AQL of 2.5 percent defective shall be used for the inspection specified in table V.

4.4 Test conditions.

4.4.1 Standard conditions. Unless otherwise specified herein, all tests shall be conducted at 24° ± 3°C (75° ± 5°F) and a relative humidity of 50 ± 4 percent.

4.4.2 Specimen conditioning. Unless otherwise specified, all test specimens shall be conditioned at standard conditions for at least 24 hours prior to undergoing tests.

4.5 Test methods.

4.5.1 ASTM methods. ASTM methods are as specified in table IV when there are no exceptions to the method. Otherwise, the ASTM procedure is identified as part of the applicable test method.

4.5.2 As received test procedures.

4.5.2.1 Hardness. Hardness shall be determined in accordance with ASTM D 2240, except that 10 readings shall be obtained on each specimen.

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4.5.2.2 Compression set. Compression set shall be determined in accordance with ASTM D 395, method B and the following:

a. Specimens die cut as specified in note 4 - ASTM D 395.
b. Type I sheet thickness shall be 0.50 inch (12.7 mm). All type I thicknesses shall not vary more than ± 0.005 inch (0.127 mm) from each other.
c. Type II sheet thickness shall be 0.25 inch (6.35mm). All type II sheet thicknesses shall not vary more than ± 0.025 inch (0.635 mm) from each other.
d. Percent deflection as specified in table VI.
e. Test conditions are $70^{\circ} \pm 1^{\circ}C$ ($158^{\circ} \pm 2^{\circ}F$) for 22 hours.
f. Results shall agree within 5 percent for each pair of specimens.

4.5.2.3 Compressibility and recovery. Compressibility and recovery shall be determined on a single layer specimen having a minimum circular or square surface area on one side of two square inches (12.9 sq cm). A compressibility testing apparatus, figure 1, or equal, with a circular presser foot having an area of 1 ± 0.001 square inch (6.45 sq cm) shall be used.

4.5.2.3.1 Procedure. The specimen shall be inserted in the test apparatus and a preload of 1 psi (6.89 KPa) applied for 15 seconds. The original thickness (A) shall be measured at this time to the nearest 0.001 inch (0.025 mm). Immediately following this, the major load specified in table VII shall be applied to the specimen, without impact, at a uniform rate, so that the load is applied within 10 ± 1 seconds. The total load (table VII) shall be maintained for a period of 60 ± 1 seconds at which time the compressed thickness (B) shall be measured to the nearest 0.001 inch (0.025 mm). Immediately remove the major load after this measurement. The specimen shall be permitted to recover for 60 ± 1 seconds under the 1 psi (6.89 KPa) preload, then the recovered thickness (C) shall be measured. Compressibility and recovery shall be calculated as follows:

$$\text{Compressibility (\%)} = \frac{A - B}{A} \times 100$$

$$\text{Recovery, \%} = \frac{C}{A} \times 100$$

where:

A = Original thickness under preload (inch, mm).

B = Thickness when compressed (inch, mm).

C = Thickness under preload after release of compressive load (inch, mm).

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4.5.3 Oven aging. Gasket sheeting material undergoing heat aging shall be subject to a temperature of 100° ± 1°C (212° ± 2°F) for a period of 70 hours ± 15 minutes using a circulating air oven, equipped so the specimens may be suspended vertically. Specimens for hardness, weight change and flexibility shall be exposed, then tested as specified below between 16 and 72 hours after removal from oven.

4.5.3.1 Hardness change. Hardness shall be determined as specified in 4.5.2.1. Change in hardness shall be determined using the obtained value and the as received hardness.

4.5.3.2 Weight change. Specimens approximately 1 by 2 by 0.063 inch (25 by 50 by 1.6 mm) shall be weighed (W_1) to the nearest 0.001 gram, exposed as specified in 4.5.3, cooled and reweighed (W_2) to the nearest 0.001 gram. Percent change in weight shall be calculated as follows:

$$\text{Weight change, } \% = \frac{W_2 - W_1}{W_1} \times 100$$

where: W_1 = initial weight.

W_2 = weight after oven aging.

4.5.3.3 Flexibility. Six - 1 by 6 inches (2.5 by 14 cm) specimens (3 from each direction) shall be oven aged (4.5.3). Upon completion of the aging cycle, each specimen shall be bent through 180° on a mandrel whose diameter is equal to 10 times the thickness of the material.

4.5.4 Immersion tests.

4.5.4.1 Procedure. Immersion media and testing conditions shall be as specified in table VIII. Changes in hardness (4.5.3.1), volume (4.5.1, table IV) and compressibility (4.5.2.3) shall be determined after immersion in oil or fuel. Change in volume (ASTM D 471) shall be determined after immersion in water.

4.5.5 Low temperature tests.

4.5.5.1 Compressibility. Compressibility at low temperature shall be determined by using the apparatus and following the procedures described in 4.5.2.3, except that the test specimen, after determination of the original thickness (A), shall be conditioned at -40° ± 1°C (-40° ± 2°F) for 5 hours ± 15 minutes before adding the major load. The major load shall be added at the low temperature. Compressed thickness (B) shall be determined as specified in 4.5.2.3. Compressibility at low temperature shall be calculated as follows:

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$$\text{Compressibility at } -40^{\circ}C, \% = \frac{A - B}{A} \times 100$$

where: A = Original thickness obtained at standard conditions.

B = Compressed thickness at $-40^{\circ}C$.

Using the "as received" compressibility value, low temperature compressibility factor shall be calculated as follows:

$$\text{Low temperature compressibility, } \% = \frac{C}{D} \times 100$$

where: C = percent compressibility at $-40^{\circ}C$.

D = percent compressibility as received.

5. PACKAGING

5.1 Preservation - packaging. Preservation - packaging shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The sheet gasket material shall be separated with paper or other suitable separator sheets which will not adhere to, or otherwise damage the material. Rolled material and sheets shall be wrapped in barrier material conforming to MIL-B-121 and sealed with tape comforming to PPP-T-76.

5.1.2 Commercial. The gasket material shall be packaged in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2).

5.2.1 Level A. The packaged gasket sheeting shall be packed in an overseas, exterior type box conforming to PPP-B-576, PPP-B-591, PPP-B-601, PPP-B-621 or PPP-B-636. Boxes shall be strapped in accordance with the appendix of the applicable box specification.

5.2.2 Level B. The gasket material shall be packed as specified in 5.2.1, except that the boxes shall be domestic type.

5.2.3 Commercial. The gasket material shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

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5.3 Marking for shipments. In addition to any special markings required by the contract (see 6.2), each package and shipping container shall be marked in accordance with MIL-STD-129 and shall include the following:

MIL-C-6183B.
National stock number.
Type, class and grade.
Manufacturer's part no. (when applicable).
Quarter and calendar year of manufacture.
Contract number.

6. NOTES

6.1 Intended use. The sheet gasket material covered by this specification is intended for use as gaskets where aromatic fuel or oil resistance is required.

6.1.1 Manufacturing processes. The choice of type I or type II is at the option of the acquiring activity. Consideration should be given to the property differences noted in table I.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, grade and form (see 1.2 and 3.3).
c. Thickness and other dimensions (see 3.5.1).
d. Manufacturer's part number (when applicable).
e. Total quantity desired.
f. Selection of applicable levels of preservation, packaging and packing (see section 5).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MR
Navy - AS
Air Force - 11

Review activities:
Army - AR, EA
Navy - YD

User activities:
Army - ME

Preparing activity:
Navy - AS
Project No. 5330-0525

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TABLE I. Physical properties (see 3.4.1).

TYPE	CLASS	GRADE	AS RECEIVED					AFTER OVEN AGING			AFTER IMMERSION							LOW TEMPERATURE COMPRESSIBILITY
											WATER	OIL			AROMATIC FUEL			
								70 ± 1/4 hr. at 100° ± 1°C (212° ± 2°F)			1 hr. ± 5 min. at 100° ± 1°C (212° ± 2°F)	70 ± 1/4 hr. at 100° ± 1°C (212° ± 2°F)			24 ± 1/4 hr. at 23° ± 1°C (73.5° ± 2°F)			5 ± 1/4 hr. at -40° ± 1°C (-40° ± 2°F)
								CHANGE IN			CHANGE IN	CHANGE IN			CHANGE IN			
			HARDNESS pts	SPECIFIC GRAVITY max	COMPRESSION SET %, max	COMPRESSIBILITY %	RECOVERY %, min	HARD pts	Wt. %, max	FLEX 180° BEND	VOLUME %	HARD. pts	VOLUME %	COMPRESSIBILITY	HARD. pts.	VOLUME %	COMPRESSIBILITY	%, min
I	AROMATIC FUEL RESISTANT	SOFT	1/	0.90	80	35 ± 5	95	+15-0	-10	2/	+15-10	±15	±15	3/	+0-20	+20	3/	10
		MEDIUM	1/	0.90	80	25 ± 5	95	+10-0	-10	2/	+15-10	±10	±15	3/	+0-20	+20	3/	10
		FIRM	1/	1.05	55	15 ± 5	95	+10-0	-10	2/	+15-10	±10	±15	3/	+0-20	+20	3/	25
	OIL RESISTANT	SOFT	1/	0.90	80	35 ± 5	95	+15-0	-10	2/	+15-10	±10	+10-0	3/	+0-20	+60	3/	25
		MEDIUM	1/	0.90	80	25 ± 5	95	+10-0	10	2/	+15-10	±10	+10-0	3/	+0-20	+60	3/	25
		FIRM	1/	1.05	55	15 ± 5	95	+10-0	-10	2/	+15-10	±10	+10-0	3/	+0-20	+60	3/	25
II	AROMATIC FUEL RESISTANT	SOFT	1/	0.90	80	25 ± 5	90	+15-0	10	2/	+15-10	±15	±15	3/	+0-30	+25	3/	15
		MEDIUM	1/	0.90	80	25 ± 5	90	+10-0	0	2/	+15-10	±10	±15	3/	+0-30	+25	3/	15
		FIRM	1/	1.30	55	15 ± 5	95	+10-0	-10	2/	+15-10	±10	±15	3/	+0-30	+25	3/	15
	OIL RESISTANT	SOFT	1/	0.90	80	35 ± 5	90	+15-0	-10	2/	+15-10	±10	+10-0	3/	+0-30	+60	3/	30
		MEDIUM	1/	0.90	80	25 ± 5	90	+10-0	-10	2/	+15-10	±10	+10-0	3/	+0-30	+60	3/	30
		FIRM	1/	1.30	55	15 ± 5	95	+10-0	-10	2/	+15-10	±10	+10-0	3/	+0-30	+60	3/	30

1/ As determined - This value shall be used to calculate change in hardness after oven aging, oil immersion, and fuel immersion.

2/ No cracking

3/ No rupture

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TABLE II. Thickness and width dimensioning (see 3.5.1).

Nominal thickness, inch (mm)	Maximum width inch (cm)
0.0157 (0.4)	18 (45.72)
0.0313 (0.8)	36 (91.44)
0.0625 (1.58)	36 (91.44)
0.0938 (2.38)	36 (91.44)
0.125 (3.18)	36 (91.44)
0.1875 (4.76)	36 (91.44)
0.25 (6.35)	36 (91.44)

TABLE III. Visual inspection (see 4.3.3.1).

Examination	Defect
Appearance	Not uniform in finish or firmness. Dirt, surface or imbedded foreign material.
Markings	Not in accordance with specification (see 3.7).
Dimensions	Any measurement not within specified dimension and tolerance as specified in 3.5, applicable drawing or contract.

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TABLE IV. Physical properties (see 3.5.2.1 and 4.3.3.2.1). 1/

Property	Test para.	No. of specimens per sample unit	Results reported to the nearest 2/
As received:			
Hardness	4.5.2.1	2	1 unit
Thickness	ASTM F 104	2	0.001 inch (0.025 mm)
Specific gravity	ASTM F 104	2	0.001 unit
Compression set	4.5.2.2	2	0.1%
Compressibility & recovery 3/	4.5.2.3	3	0.1%
After oven aging:	4.5.3		
Change in hardness, pts	4.5.3.1	2	1 unit
Change in weight, %, max	4.5.3.2	2	0.1%
Flexibility	4.5.3.3	6	pass/fail
After immersion:	4.5.4		
In water:	Table VIII		
Change in volume, %	ASTM D 471	3	0.1%
In oil:	Table VIII		
Change in hardness, pts	4.5.4.1	2	1 unit
Change in volume, %	ASTM D 471	3	0.1%
Compressibility	4.5.4.1	3	0.1%
In aromatic fuel:	Table VIII		
Change in hardness, pts	4.5.4.1	2	1 unit
Change in volume, % 3/	ASTM D 471	3	0.1%
Compressibility	4.5.4.1	3	0.1%
Low temperature property	4.5.5		
Compressibility	4.5.5.1	3	0.1%

1/ Requirement paragraph is table I, except for thickness, which is 3.5.2.1. All properties are to be tested for first time deliveries. Only those identified by footnote 3/ for subsequent deliveries.

2/ Average value shall be reported. All obtained data shall be listed.

3/ Unless otherwise specified by the acquiring activity, these properties are for subsequent acquisitions.

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TABLE V. Packaging inspection (see 4.3.3.3).

Examination	Defect
Packaging, packing	Not level specified in contract or order. Construction not as specified. Incomplete closure, missing components.
Count	Not the required count per package or pack.
Marking	Omitted, illegible, incomplete or incorrect. Not in accordance with contract or order.

TABLE VI. Deflections for compression set (see 4.5.2.3).

Grade	Deflection (% of orig. thickness)
Soft	30
Medium	25
Firm	20

TABLE VII. Compressibiity and recovery test loads (see 4.5.2.3.1).

Grade	Preload psi (KPa)	Major load psi (KPa)	Total load psi (KPa)
Soft	1 (6.89)	199 (1372)	200 (1379)
Medium	1 (6.89)	299 (2061)	300 (2068)
Firm	1 (6.89)	299 (2061)	300 (2068)

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TABLE VIII. Immersion fluids and conditions (see 4.5.4.1).

Fluid	Conforming to	Immersion temp.	Duration
Distilled water	--	100° ± 1°C (212° ± 2°F)	1 hr. ± 5 min
Oil (ASTM #1)	ASTM D 471	100° ± 1°C (212° ± 2°F)	70 hrs. ± 15 min
Aromatic fuel	Type II, TT-S-735	23° ± 1°C (73.4° ± 2°F)	24 hrs. ± 15 min

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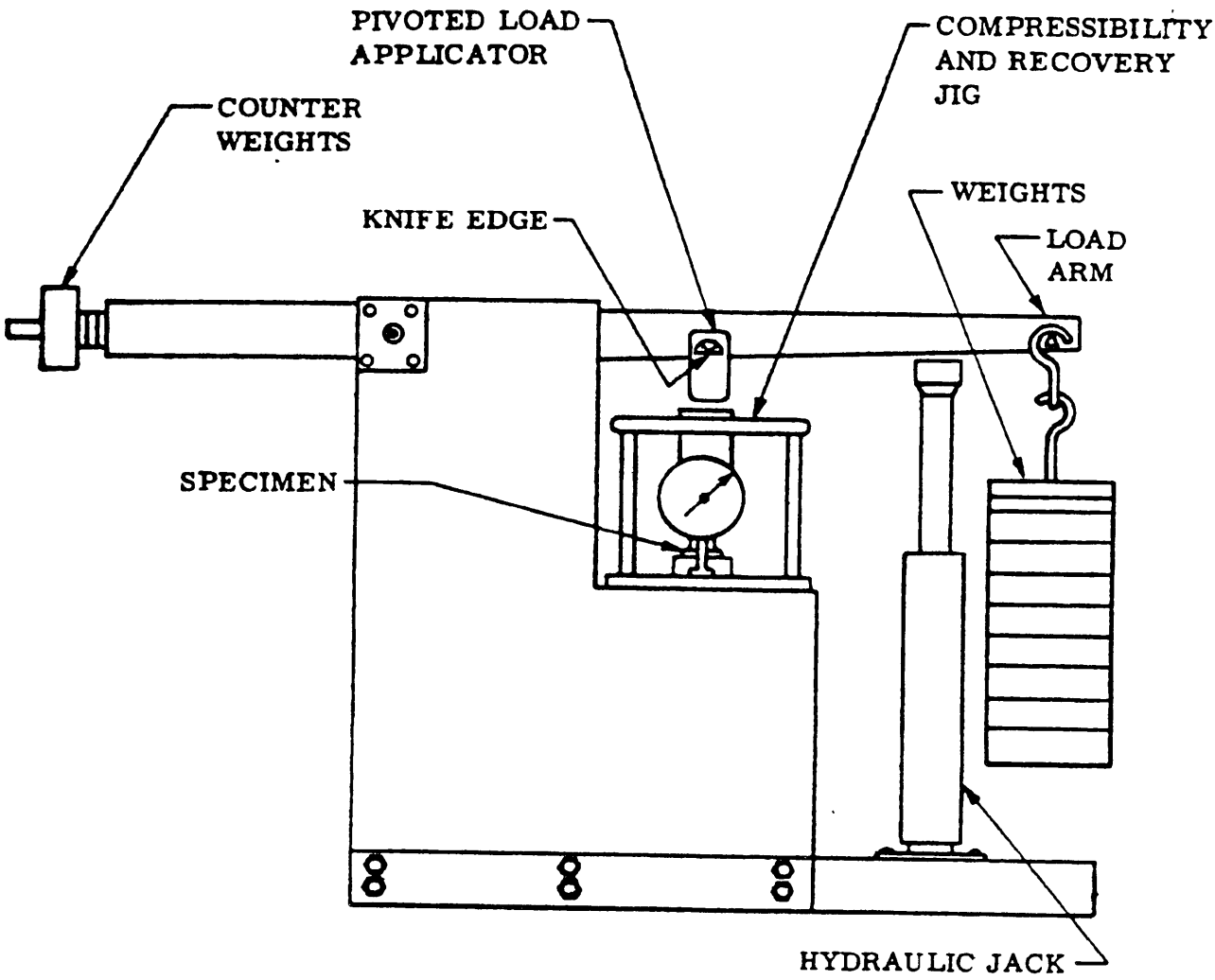


FIGURE 1. Compressibility testing machine.

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